Ferroelectric HfO_2 Memory — Ontology Specification

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- Every class, property and individual carries a source tag in the form `S1{I A}` = source S1, Section I A.
- ⇒ **UNCERTAIN** marks something that is contested between sources, implied rather than stated, out of context, or otherwise not safe to assert.
- × **INFERRED** marks modelling structure I introduced that no source states. These are design decisions, not findings.
- Anything without a tag is ontology scaffolding, not a claim about the world.

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1 Modelling conventions

These decisions govern the whole specification. They are stated here so the OWL file can be checked against a written rule rather than have its conventions reverse-engineered from it.

1.1 Naming

Classes are `Capitalised_With_Underscores`; properties are `camelCase`; unit-bearing data properties carry the unit as a suffix (`hasProcessTemperature_C`). Individuals follow the class convention, with flow-step individuals prefixed by their flow letter and position (`B1_5_HZO_ALD`). This is the same rule used in the SMT2020 ontology, so both artifacts are checkable against one convention.

1.2 Annotation properties

The tagging scheme used throughout this document is typographic here. In OWL it must become explicit annotation properties, declared before anything else, since they carry what distinguishes this ontology from a bare class tree.

Annotation property	Value	Purpose
`sourceLocator`	xsd:string, e.g. `S1{II A 2}`	The per-entity source tag. Repeatable: an entity may be attested by several sources
`epistemicStatus`	{ `verified`, `uncertain`, `inferred`, `scaffolding`, `named_only` }	Formalises the ⇒ **UNCERTAIN** / × **INFERRED** markers and the untagged-scaffolding rule from the front matter
`uncertaintyRef`	xsd:string, e.g. `U13`	Points an entity at its entry in the register in §8.2, so the register is reachable from the model rather than only from this document
`contradictedBy`	xsd:string, e.g. `S1`	Marks an entity whose own source contradicts itself. Needed by `A6_Patterned_Metal_Deposition`; see the instance document
`scopeNote`	xsd:string	Records what a class deliberately excludes

1.3 Measured versus specified

Two kinds of number appear in the sources and they must not be modelled the same way.

Specified values are set, targeted or nominal: process setpoints, as-designed thicknesses, nominal compositions, geometry. These are data properties on the entity they configure — `Unit_Process`, `Layer_Slot`, `Memory_Cell_Architecture`.

Measured values are the outcome of a characterisation on a specimen under conditions. These are `Measurement` instances, never data properties on `Material`. A measured value stored on a material silently asserts that the value belongs to the substance rather than to one film on one wafer measured one way, which is exactly the failure §4.1 exists to prevent.

Where a source gives both a nominal and a measured figure for the same quantity — HZO's 1:1 nominal against S24's measured 0.8 — the nominal is a data property and the measured figure is a `Measurement`. They are different claims.

1.4 Disjointness

Asserted where the sources support it, declined where they do not, and the declining is recorded rather than left to a default.

Disjoint: the top-level siblings `Material`, `Dopant`, `Phase_State`, `Defect`, `Layer_Stack`, `Layer_Slot`, `Memory_Cell_Architecture`, `Array_Architecture`, `Process_Flow`, `Unit_Process`, `Equipment`, `Capability`, `Measurement_Method`, `Measurement`, `Measured_Property`, `Artifact`, `Technology_Node`, `Degradation_Phenomenon`. Also the crystalline phases, which are mutually exclusive space groups.

Deliberately not disjoint:

- `Substrate` and `Channel_Material`. Silicon is both, in different flows. The distinction is a role, and roles live on `Layer_Slot`.
- The subclasses of `Degradation_Phenomenon`. A real film shows several at once; this is the same argument already made for the causal properties in §3.4.
- `Ferroelectric_Material` and `Dielectric_Layer_Material`. HfO_2 is dielectric in one stack position and ferroelectric in another depending on phase.

1.5 Open-world reading

`rdfs:domain` and `rdfs:range` are inference rules, not constraints: asserting a property on an unexpected individual reclassifies it silently rather than raising an error. Nothing in this specification should be read as validation. If closed-world checking is wanted — every `Unit_Process` must have a `belongsToFlow`, for instance — that is SHACL's job and belongs in a separate shapes file. The absences recorded in §8.3 depend on this reading being preserved.

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2 Classes

Indentation = `rdfs:subClassOf`. Curly braces list verified individuals.

```
Thing
|
+-- Material
|   |
|   +-- Substrate
|   |     { Si_wafer, Si_001, Si_p_doped_heavy, Si_3inch_10ohmcm,
|   |       Si_8inch_p_type, SiO2_on_Si_PECVD }                    S1{II}, S9, S11, S23, S24, S27
|   |     [UNCERTAIN] Epitaxial substrates (YSZ, LSMO, ITO, scandates) appear only in
|   |       S1{II C} as PLD research substrates, not production   S1{II C}
```

```
|   |
|   +-- Ferroelectric_Material
|   |     [SCOPE] Every subclass is HfO2-family. Perovskite and other non-HfO2
|   |       ferroelectrics are out of scope by design, not absent for lack of
|   |       evidence. The genus keeps its general name so the scope can be
|   |       widened later without renaming; the exclusion is recorded as a
|   |       scopeNote annotation, not baked into the class name.
|   |   +-- Undoped_HfO2                  { HfO2 }                    S1{I B}
|   |   +-- Doped_HfO2
|   |   |     { Si_HfO2, Y_HfO2, Sr_HfO2, La_HfO2, Gd_HfO2, Al_HfO2,
|   |   |       N_HfO2, Fe_HfO2, Sc_HfO2, Ge_HfO2 }                   S1{I B, II B 1}
|   |   +-- Hafnium_Zirconate         { HZO }                         S1{I}
|   |   |     [UNCERTAIN] "HZO" is a composition range, not a fixed formula.
|   |   |       S24 measured Hf:Zr ~ 0.8, i.e. Hf0.45Zr0.55O2, and still
|   |   |       calls it HZO. Use hasHfZrRatio, do not infer 50:50.      S24{2}
|   |   +-- Doped_Hafnium_Zirconate   { La_HZO, Al_codoped_HZO }      S27, S6
|   |   +-- Hafnium_Aluminate         { HAO }  Al:Hf = 1:30           S9{p.2}
|   |
|   +-- Electrode_Material
|   |     { TiN, TaN, Ru, W, Pt, IrO2, Al }                           S1{II A 2}, S12a, S24
|   |     [UNCERTAIN] In the polycrystalline MFM literature Ti/Pt is a contact pad,
|   |       but in PLD and CSD literature Pt is a functional electrode   S1{II}
|   |
|   +-- Dielectric_Layer_Material
|   |   +-- Interfacial_Layer_Material  { SiO2, SiOx, SiON }          S9, S10, S11
|   |   +-- Insertion_Layer_Material    { Al2O3 }                     S1{II A 3}, S11
|   |   +-- Seed_Layer_Material         { HfO2_seed, ZrO2_seed }      S1{II A 3}
|   |   +-- Interface_Reaction_Product  { TiO2, HfN }                 S1{II B 2}
|   |         TiO2 forms deliberately (pre-deposition O2) or parasitically
|   |         (excess O3). HfN is the defect this prevents.
|   |
|   +-- Channel_Material
|   |     { Si_bulk, polySi, IGZO }                                   S5, S6
|   |     [UNCERTAIN] ITO and IWO named by S6 as a class of oxide semiconductors
|   |       under investigation; no device data verified              S6
|   |
|   +-- Chemical_Precursor
|   |   +-- Hf_Precursor      { HfCl4, TDMA_Hf, TEMA_Hf, CpHf_NMe2_3 }   S1{II A 1}
|   |   +-- Zr_Precursor      { TDMA_Zr, TEMA_Zr, CpZr_NMe2_3 }         S1{II A 1}
|   |   +-- Dopant_Precursor                                          S1{Table I}
|   |   |     [UNCERTAIN] Individual dopant precursors are listed in S1 Table I but
|   |   |       none was verified in a device flow. Treat as catalogue.
|   |   +-- Electrode_Precursor { TiCl4, NH3, Ru_EtCp_2 }              S23{2}, S24{2}
|   |
|   +-- Oxidant             { H2O, O3, O2_gas, O2_plasma, H2O2 }        S1{II A 2}, S23{2}
|   |
|   +-- Process_Gas
|   |     { N2, Ar, O2, NH3, forming_gas_5H2_95N2, SF6 }               S1, S9, S10, S24
|   |
|   +-- Sputter_Target
|   |     { HfO2_target, ZrO2_target, Hf_metal, Y2O3_target, HZO_target } S1{II B, Table III}
|   |
|   +-- Wet_Chemical        { SC1, dilute_HF }                          S1{II}, S10, S11
|
+-- Dopant                  { Si4, Y3, Sr2, La3, Gd3, Al3, N3, Zr4,
|                             Fe3, Sc, Ge, Ce, Nd, Sm, Er, Yb }         S1{I B, II B 1, II D}
|
+-- Phase_State
|   +-- Amorphous                                                     S1{II A 1}
|   +-- Crystalline_Phase
|       +-- Orthorhombic_Pca21          ferroelectric                 S1{I A}
|       +-- Orthorhombic_Pbca           non-ferroelectric             S1{II E}
|       +-- Tetragonal_P42nmc           non-ferroelectric             S1{I A}
```

```
|       +-- Monoclinic_P21c             non-ferroelectric, bulk-stable   S1{I A}
|       +-- Cubic_Fm3m                                                   S1{I A}
|
+-- Defect
|   +-- Point_Defect
|   |   +-- Oxygen_Vacancy       stabilises o- and t-phase               S1{I B, II A 2}
|   |   +-- Oxygen_Interstitial  removes barrier to m-phase              S1{II A 2}
|   |   +-- Metal_Vacancy        Hf, Zr -- origin of deep traps           S10{III D}
|   +-- Electronic_Trap          { Et1, Et2 }                            S10{III D}
|
+-- Layer_Stack
|   +-- MFM        metal-ferroelectric-metal                             S1{II}
|   +-- MFIS       metal-ferroelectric-insulator-semiconductor           S9, S10, S11
|   +-- MIFIS      metal-insulator-ferroelectric-insulator-semiconductor S11{II}
|   +-- MFMIS      metal-ferroelectric-metal-insulator-semiconductor     S5, S6
|   +-- Nanolaminate                                                     S1{II A 3}
|   +-- Bilayer                                                          S1{II A 3}
|   +-- Trilayer                                                         S1{II A 3}
|   +-- Heterogeneous_Co_Doped_Stack   e.g. IL/HAO/HAO/HZO               S9{p.2}
|
+-- Layer_Slot                           [INFERRED] REIFIED. One position in one stack.
|     Carries layerIndex, the material occupying it, the role it plays there,
|     and the as-fabricated thickness. See S3.3.
|
+-- Layer_Role
|     { substrate, interfacial_layer, insertion_layer, seed_layer,
|       bottom_electrode, ferroelectric_layer, internal_electrode,
|       top_electrode, gate_metal, contact_pad, channel }       [INFERRED]
|
+-- Artifact                             [INFERRED] the physical specimen measured
|   +-- Blanket_Film                     unpatterned film on a wafer     S1, S9, S27
|   +-- Patterned_Capacitor              MFM / MIM test structure        S12a, S23, S24
|   +-- Single_Cell                      one FeFET or one 1T1C cell      S10, S11, S12a
|   +-- Array_Specimen                   an array measured as a unit     S12a
|         S12a measures a 1C capacitor and a 1T-1C array from the same wafer
|         and reports they agree. Two artifacts, one wafer.               S12a{P5}
|
+-- Technology_Node                      { node_130nm }                  S12a{P4}
|
+-- Memory_Cell_Architecture
|   +-- Capacitor_Based_Cell
|   |   +-- FeCAP_1C                                                     S12a{Fig. 1b}
|   |   +-- FeRAM_1T1C_Charge_Sensed     destructive read                S5
|   |   +-- FeRAM_1T1C_Capacitance_Sensed non-destructive, <2x ratio      S5
|   +-- Transistor_Based_Cell
|   |   +-- FeFET_1T                                                     S5
|   |   +-- FeMFET_1T1C                  MFMIS                           S5
|   +-- Hybrid_Cell
|   |   +-- FeRAM_2T1C                                                   S5
|   |   +-- FeRAM_2TnC                   quasi-non-destructive readout   S5
|   |   +-- FeRAM_1TnC                                                   S5
|   +-- FTJ                             epistemicStatus = named_only; uncertaintyRef = U19
|         [UNCERTAIN] NAMED ONLY. Listed as an application of ferroelectric HfO2
|           by S1{I} and S27{p.1134}. No structure, sensing mechanism or
|           performance data verified. Do not populate.
|
+-- Array_Architecture
|   +-- VC_FeNAND     vertical channel                                   S5
|   +-- HC_FeNAND     horizontal channel                                 S5
|   +-- Fe_AND                                                           S5
|   +-- Stacking_Strategy
|       +-- Parallel_Stacking    lithography count ~independent of layers S5
|       +-- Sequential_Stacking  lithography scales ~linearly with layers S5
```

```
|
+-- Process_Flow
|     { FlowA_LabMFM, FlowB1_GateLastFeFET, FlowB2_GateLastFeFET_MIFIS,
|       FlowC_BEOL_FeRAM, FlowD_MFIS_Capacitor, FlowE_FullALD_AutoAnneal,
|       FlowF_MFM, FlowG_LaHZO_MIM }                                     S4
|
+-- Unit_Process
|   +-- Substrate_Preparation
|   |   +-- Wet_Clean                    { RCA_SC1, dHF_clean }          S1, S10, S11
|   |   +-- Native_Oxide_Removal                                         S9{p.2}
|   |   +-- Pre_Deposition_Oxygen_Flow   forms self-limiting TiO2        S1{II B 2}
|   +-- Interfacial_Layer_Formation
|   |   +-- Ozone_Oxidation              O3, 300 C, in ALD chamber       S10{II A}, S11{II}
|   |   +-- Chemical_Oxidation_Plus_Nitridation   RTP in NH3, then O2/N2 S9{p.2}
|   +-- Deposition
|   |   +-- ALD
|   |   |   +-- Thermal_ALD                                              S24{2}
|   |   |   +-- Plasma_Enhanced_ALD                                      S23{2}
|   |   |   +-- Radical_Enhanced_ALD     Ru from Ru(EtCp)2 + O*          S24{2}
|   |   +-- PVD
|   |   |   +-- Sputter_Deposition       { DC, RF, magnetron, reactive } S1{II B}
|   |   |   +-- Evaporation                                              S1{II}
|   |   +-- PLD                                                          S1{II C}
|   |   +-- CSD                          { spin_coat, dip_coat }         S1{II D}
|   |   +-- PECVD                        SiO2 insulation                 S24{2}
|   +-- Thermal_Process
|   |   +-- Crystallisation_Anneal       RTA/RTP                         S1{II}, S9-S12a, S23
|   |   +-- Post_Metallisation_Anneal    furnace, 1 h                    S27{p.1134}
|   |   +-- Dopant_Activation_Anneal     1050 C / 5 s                    S10, S11
|   |   +-- Forming_Gas_Anneal           450 C, 5% H2/95% N2             S10, S11
|   |   +-- Retention_Bake               85 C diagnostic retention       S24{3.3}
|   |         105 C fatigue recovery test                                S24{3.2}
|   +-- Implantation                     channel doping; S/D formation   S10, S11
|   +-- Lithography                      { photolithography, hard_mask } S1{II}, S24{2}
|   +-- Etch
|   |   +-- Wet_Etch                     { SC1_TiN_removal }             S1{II}
|   |   +-- Plasma_Etch                  { SF6_Ar_TiN, O2_Ar_Ru }        S24{2}
|   +-- BEOLProcess
|   |   +-- Isolation_Deposition                                         S12a{Fig. 1a}
|   |   +-- Via_Patterning_And_Etch                                      S12a{Fig. 1a}
|   |   +-- Metal_Deposition_And_Patterning                              S12a{Fig. 1a}
|   +-- Contact_Formation                { TiN_Al, Al_pads }             S10, S24
|   +-- Characterisation                 [INFERRED] a metrology or electrical test
|   |     step, so that characterisation sits inside the flow ordering and can
|   |     use a Metrology_Tool or Electrical_Test_Tool through usesEquipment.
|   |     Without it, those tools are unreachable: usesEquipment has domain
|   |     Unit_Process, and no Unit_Process subclass performed measurement.
|   |     Produces Measurement instances via producesMeasurement.
|   |   +-- Structural_Characterisation   GIXRD, TEM, PED, ToF-SIMS       S9, S24, S27
|   |   +-- Electrical_Characterisation   P-E, PUND, NDPU, IdVg, C-V      S10, S11, S23, S24
|   +-- Electrical_Conditioning
|       +-- Wake_Up_Cycling              10^5 cycles, 3 us, 3 V          S24{3.2}
|             [UNCERTAIN] Verified as an experimental preparation step, not as a
|               production process step. v1 claimed the latter on an
|               unobtainable source. Model as a conditioning step used
|               before characterisation; do not assert it is in any
|               manufacturing flow.
|
+-- Equipment
|   +-- Deposition_Tool
|   |     { ALD_reactor, PEALD_reactor, REALD_reactor,
|   |       DC_magnetron_sputter, RF_magnetron_sputter, PLD_chamber,
|   |       PECVD_tool, spin_coater }                                    S1, S23, S24
```

```
|   |     Verified instance: Picosun R200adv                             S24{2}
|   +-- Thermal_Tool
|   |     { RTP_system, furnace }                                        S1{II}, S9, S27
|   |     [INFERRED] v1 listed a UV nanosecond laser anneal tool from an
|   |       unverified source. Removed.
|   +-- Lithography_Tool                                                 S1{II}, S24{2}
|   |     [UNCERTAIN] No verified source names a specific exposure tool. v1's
|   |       "DUV scanner vs mask aligner" distinction is unsupported.
|   +-- Etch_Tool                      plasma etcher                     S24{2}
|   +-- Implant_Tool                   ion implanter                     S10, S11
|   +-- Metrology_Tool
|   |     { GIXRD_diffractometer, TEM, HRTEM, PED_ASTAR_system,
|   |       ToF_SIMS, XPS, EDS }                                         S1, S9, S24, S27
|   |     Verified instances: ARL X'TRA (GIXRD) S24{2};
|   |       Thermo Fisher Titan3 G2 60-300 + Nanomegas DIGISTAR P2010
|   |       with Topspin/ASTAR + DensSolutions heating-biasing holder    S27{p.1134}
|   +-- Electrical_Test_Tool
|         { probe_station, ferroelectric_tester, parameter_analyzer,
|           pulse_generator }                                            S10, S24, S23
|         Verified instances: Keysight/Agilent B1500A S10{II A}, S11{II},
|           S24{2}; Radiant Precision LC S10{II A};
|           Summit 11000B-M probe station S24{2};
|           F3000, Zhejiang Liryder S23{2}
|
+-- Capability
|     { Conformality, SubNanometreThicknessControl, LineOfSight,
|       SelfLimitingGrowth, HighDepositionRate, LowCarbonContamination,
|       EtchSelectivityToHfO2, EpitaxialGrowth }                         S1{II A, II B, III}, S24{1}
|
+-- Measurement_Method
|   +-- PE_Hysteresis                                                    S10, S24
|   +-- PUND                                                             S9, S23, S24
|   +-- NDPU                                                             S23
|   +-- Dynamic_IswE                                                     S24{2}
|   +-- Small_Signal_CV                                                  S24{2}
|   +-- QSCV                                                             S10{II C}
|   +-- Endurance_Cycling                                                S9-S12a, S24
|   +-- Retention_Measurement   SS / NSS / OS states                     S24{3.3}
|   +-- IdVg_Transfer                                                    S10, S11
|   +-- Permittivity_From_Displacement_Current                           S23{3.2}
|   +-- GIXRD_Phase_Analysis                                             S9, S24
|   +-- PED_Orientation_Mapping                                          S27
|   +-- ToF_SIMS_Depth_Profiling                                         S9
|
+-- Measurement                        [INFERRED] REIFIED n-ary class. See S3.1
|
+-- Measured_Property
|   +-- Polarisation_Property
|   |     { remanent_polarisation_Pr, double_remanent_2Pr,
|   |       switched_polarisation_Psw, saturated_polarisation_Ps }
|   +-- Field_Property
|   |     { coercive_field_Ec, coercive_voltage_Vc, bias_field_Ebias,
|   |       depolarisation_field_Ed, interfacial_field_EI,
|   |       built_in_field_Ebuilt_in, breakdown_field }
|   +-- Device_Property
|   |     { memory_window_MW, threshold_voltage_Vth, sense_margin,
|   |       bitline_voltage_VBL, switching_speed, on_current_Ion }
|   +-- Dielectric_Property
|   |     { relative_permittivity_k, leakage_current_density }
|   +-- Reliability_Property
|   |     { endurance_cycles, retention_time, imprint_shift,
|   |       wakeup_magnitude, trapped_charge_density }
|   +-- Structural_Property
```

```
|        { phase_fraction, grain_size, grain_aspect, film_thickness,
|          carbon_impurity, oxygen_vacancy_concentration,
|          growth_per_cycle, film_density, crystallographic_texture }
|
+-- Degradation_Phenomenon
    |     [INFERRED] The two-way split below is a modelling decision. It exists
    |       because the two groups attach to different things: the film-level
    |       phenomena are properties of a Layer_Stack, the array-level ones only
    |       exist once cells are wired into an Array_Architecture. Before the
    |       split, exhibits had no domain that could carry Pass_Disturb.
    +-- Film_Level_Degradation
    |   +-- Wake_Up
    |   |   +-- First_Stage_Wake_Up                             S24{3.2}
    |   |   +-- Second_Stage_Wake_Up activates >343 K at 10^7-10^8 cycles S24{3.2}
    |   +-- Wake_Up_Relaxation   Vo driven back during idle at 0 V       S23{4}
    |   +-- Fatigue                                             S24{3.2}
    |   +-- Imprint                                             S23, S24
    |   |     [UNCERTAIN] "Fluid imprint" reported by S6 as a distinct rapid
    |   |       mechanism under repeated non-destructive read.
    |   |       Trade-press only.                               S6
    |   +-- Retention_Loss
    |   |   +-- Relaxation       ruled out in S24's devices             S24{3.3}
    |   |   +-- Depolarisation_Loss                             S24{3.3}
    |   |   +-- Imprint_Driven_Loss                             S24{3.3}
    |   +-- Charge_Trapping                                     S5, S10, S11
    |   +-- Dielectric_Breakdown                                S24{3.2}
    +-- Array_Level_Degradation
        +-- Pass_Disturb         3D NAND strings                S5
        +-- Read_Disturb         accumulative switching in QNRO         S5
        [UNCERTAIN] "Localised phase variation -> Vth variation" reported by S6
          only (trade press). Not modelled as a class.
```

3 Object properties

3.1 Process ordering

Property	Domain	Range	Characteristics	Notes
`belongsToFlow`	Unit_Process	Process_Flow	—	**Required on every Unit_Process individual.** Without it, `precedes` produces contradictions between Flow C and the rest
`directlyPrecedes`	Unit_Process	Unit_Process	asymmetric, irreflexive	Immediate ordering within one flow
`precedes`	Unit_Process	Unit_Process	transitive *only*	Transitive closure of `directlyPrecedes`, **scoped by flow**. Asymmetry and irreflexivity are **not** declared: OWL 2 DL permits them only on simple properties, and a transitive property is non-simple. See §9.4

Property	Domain	Range	Characteristics	Notes
`flowProducesArchitecture`	Process_Flow	Memory_Cell_Architecture ∪ Layer_Stack	—	Flow C → FeRAM_1T1C; Flow E → MFM; Flow B1 → MFIS
`flowTargetsNode`	Process_Flow	Technology_Node	—	Flow C → 130 nm `S12a{¶4}`

Read the following at individual level. Each bullet names classes for brevity, but the assertion holds between the *step individuals* of the named flow — `A4` and `A5` in Flow A, `B1_6` and `B1_7` in Flow B1 — never between the classes themselves. `precedes` is an object property over `Unit_Process` individuals; asserting it between classes would require punning and would make the transitive closure meaningless, since Flow C orders the same two class-level steps the other way round.

Verified ordering facts:

- `directlyPrecedes(Top_Electrode_Deposition, Crystallisation_Anneal)` holds in Flows A, C, D, F, G and B1/B2. `S1, S9, S10, S11, S12a, S23, S27`
- `precedes(Crystallisation_Anneal, Patterning)` holds in **A, B1, B2, D, F, G**. (Flow E has no `Crystallisation_Anneal` individual; instead, `E5_TiN_TE_ALD` — which `transformsPhase` — precedes `E6_Patterning`.)
- `precedes(Patterning, Crystallisation_Anneal)` holds in **Flow C alone**. `S12a{Fig. 1a}`
- `precedes(Dopant_Activation_Anneal, Interfacial_Layer_Formation)` holds in the gate-last flows B1, B2. `S10, S11`
- Flow E has **no** `Crystallisation_Anneal` individual at all; crystallisation occurs during the ALD TiN top electrode deposition (E5), which is a `Unit_Process` with `transformsPhase = true`. `S24{2}`

3.2 Process composition

Property	Domain	Range	Notes
`hasInput`	Unit_Process	Material ∪ Artifact	Many-to-many
`hasOutput`	Unit_Process	Material ∪ Artifact	
`usesEquipment`	Unit_Process	Equipment	Many-to-many. One ALD reactor serves electrode and ferro-electric deposition `S24{2}` (S24 has no interfacial layer)
`usesPrecursor`	Deposition	Chemical_Precursor	
`usesOxidant`	Deposition	Oxidant	
`usesProcessGas`	Unit_Process	Process_Gas	Covers carrier, sputter, anneal ambient and etch gases
`usesTarget`	Sputter_Deposition	Sputter_Target	
`depositsMaterial`	Deposition	Material	
`etchesMaterial`	Etch	Material	`SF6_Ar_TiN etchesMaterial TiN` `S24{2}`
`hasAmbient`	Thermal_Process	Process_Gas	N_2 recommended over O_2 `S1{II B 2}`

3.3 Structure and composition

Property	Domain	Range	Notes
`hasSlot`	Layer_Stack	Layer_Slot	**Ordered, via the slot's `layerIndex`.** Replaces the earlier `hasLayer` / `hasLayerAtPosition` pair and the open choice between them; see the note below
`slotMaterial`	Layer_Slot	Material	The material occupying this position. Functional: one slot, one material
`slotRole`	Layer_Slot	Layer_Role	The function the material serves *here*. `Pt` is an `Electrode_Material` by class but a `contact_pad` by role in an MFM coupon and a `bottom_electrode` by role in a PLD stack S1{II}
`hasElectrode`	Layer_Stack	Electrode_Material	**Derived, not asserted.** A shortcut for the material of a slot whose `slotRole` is an electrode role. Sub-properties `hasTopElectrode`, `hasBottomElectrode` follow the corresponding roles. Kept for query convenience; the slot is authoritative
`hasTopElectrode`	Layer_Stack	Electrode_Material	Derived. `rdfs:subPropertyOf hasElectrode`
`hasBottomElectrode`	Layer_Stack	Electrode_Material	Derived. `rdfs:subPropertyOf hasElectrode`
`hasInterfacialLayer`	Layer_Stack	Interfacial_Layer_Material	Derived, from `slotRole = interfacial_layer`
`hasFerroelectricLayer`	Layer_Stack	Ferroelectric_Material	Derived, from `slotRole = ferroelectric_layer`
`hasChannel`	Memory_Cell_Architecture	Channel_Material	
`usesStack`	Memory_Cell_Architecture	Layer_Stack	FeFET_1T → MFIS; FeMFET_1T1C → MFMIS S5
`hasDopant`	Ferroelectric_Material	Dopant	
`hasPhase`	Material ∪ Layer_Stack	Phase_State	⇒ **UNCERTAIN** Films are phase *mixtures.*
`arrangedAs`	Memory_Cell_Architecture	Array_Architecture	
`usesStackingStrategy`	Array_Architecture	Stacking_Strategy	

Why the slot is reified. × **INFERRED** Three facts in the sources cannot be carried by a direct `Layer_Stack` → `Material` link, and all three are solved by the same move:

- **Role is positional, not substantial.** In the Flow A coupon, `Ti_Pt_contact` is an `Electrode_Material` by class but a probe pad by function; in PLD and CSD stacks Pt is a functional electrode S1{II}. Attaching the role to the slot states this once instead of excluding the material from the electrode relations by hand.
- **One material, two roles.** `Si_substrate` is a substrate in Flow A and the channel in Flow B1. With the role on the slot the material stays a single individual, and `Substrate` and `Channel_Material` need not be disjoint — see §1.4.
- **Thickness is per-stack.** HZO is 9 nm in Flow B1 and 9.5 nm in Flow G S10, S27. A `hasFilmThickness_nm` on the shared `HZO` individual would assert both of a single substance. Thickness belongs to the slot.

This is the same reification already used for `Measurement` in §4.1, for the same reason: a fact that only holds relative to a context needs the context to exist as an individual.

3.4 Causal properties

These are experimentally inferred, hold under conditions, and are multiply realised. Do **not** give them OWL characteristics such as functionality, and do not declare their causes disjoint — a real film has several at once.

Property	Domain	Range	Verified instances
`stabilisesPhase`	Material ∪ Defect ∪	Phase_State	`TiN_capping stabilises Orthorhombic_Pca21` S1{I A}; `Oxygen_Vacancy stabilises Orthorhombic_Pca21` and `Tetragonal_P42nmc` S1{I B, II A 2}; `ZrO2_alloying stabilises Orthorhombic_Pca21` S1{I B}; `TensileStrain stabilises Orthorhombic_Pca21` S1{II C}
`destabilisesPhase`	Material ∪ Defect ∪	Phase_State	`Oxygen_Interstitial destabilises Orthorhombic_Pca21` by removing the barrier to monoclinic S1{II A 2}; `ExcessOxygen stabilises Monoclinic_P21c` S1{II A 2}
`transformsPhase`	**Unit_Process**	Phase_State	**Domain is Unit_Process, not Thermal_Process.** In Flow E an ALD electrode deposition performs the transformation S24{2}
`scavengesOxygenFrom`	Electrode_Material	Ferroelectric_Material	`TiN`, `TaN` scavenge; `IrO2` does not, and degrades ferroelectricity S1{II A 2}; Ru has lower scavenging ability than TiN S24{4}
`diffusesInto`	Dopant ∪ Material	Material	`Zr4 diffusesInto SiON`, degrading the interfacial layer S9{p.5}
`templates`	Material	Material	⇒ **UNCERTAIN** t-phase grains sit within $< 5^\circ$ of both TiN layers, suggesting near-epitaxial growth during PMA. S27 says “may be some kind of” — implied, not demonstrated S27{p.1134}
`exhibits`	Layer_Stack ∪ Memory_Cell_Architecture ∪ Array_Architecture	Degradation_Phenomenon	Domain widened to include `Array_Architecture`. `Pass_Disturb` is a 3D NAND string phenomenon S5 and had no possible subject before
`mitigatedBy`	Degradation_Phenomenon	Unit_Process ∪ Material	`Fatigue mitigatedBy TiN` relative to Ru S24{3.2}; `Monoclinic_formation mitigatedBy Al2O3_insertion` S1{II A 3}
`enables`	Capability	Array_Architecture ∪ Memory_Cell_Architecture	`Conformality enables 3D_capacitor` S1{II B}
`precludes`	Capability	Array_Architecture ∪ Memory_Cell_Architecture	`LineOfSight precludes 3D_capacitor` S1{II B}. Blocking was previously stated only in prose on `enables`, which left it unassertable
`requiresCapability`	Unit_Process ∪ Array_Architecture	Capability	
`constrainsMaterialChoice`	Capability	Material	⇒ **UNCERTAIN** Ru selected partly because dry plasma etch selective to HfO_2 exists for it S24{1} — stated as a motivation, not a demonstrated constraint

3.5 Measurement and evidence

Property	Domain	Range	Notes
`hasMeasurement`	Artifact ∪ Layer_Stack ∪ Memory_Cell_Architecture	Measurement	

Property	Domain	Range	Notes
producesMeasurement	Characterisation	Measurement	Links the characterisation *step* to what it produced. A characterisation is a Unit_Process, so it also carries belongsToFlow, its position in the ordering, and its usesEquipment tool
measurementMethod	Measurement	Measurement_Method	
measuresProperty	Measurement	Measured_Property	
measuredOn	Measurement	Artifact	**Distinguish blanket film / patterned capacitor / single cell / array.** S12a measures both 1C and 1T-1C and reports they agree S12a{¶5}

---

4 Data properties

4.1 The Measurement class

Measurement is a reified n-ary relation. A bare hasEnduranceCycles on a material is misleading: S24's TiN device breaks down at 10^9 and its Ru device survives 10^{10}, at identical 3 µs / 3 V pulses; S12a reports 10^9 at array level and 10^{11} at capacitor level from the same wafer.

Per §1.3, every measured quantity in this specification is a Measurement. Phase fraction now runs through the same machinery rather than through three loose data properties on Phase_State: §8.1 already required fractionMethod and phasesLumped to travel with any fraction, which is the same value-plus-conditions shape this class exists for. phase_fraction is a Measured_Property and GIXRD_Phase_Analysis and PED_Orientation_Mapping are Measurement_Methods, so no new class is needed.

```
Measurement
  measurementMethod          -> Measurement_Method
  measuresProperty           -> Measured_Property
  measuredOn                 -> Artifact
  usesEquipment              -> Metrology_Tool | Electrical_Test_Tool
  hasMeasuredValue           xsd:double      renamed from hasValue: owl:hasValue is
                                             a reserved OWL construct and the name
                                             collides on sight even across namespaces
  hasUnit                    xsd:string
  hasAppliedVoltage_V        xsd:double
  hasAppliedField_MVcm       xsd:double
  hasFrequency_Hz            xsd:double
  hasPulseWidth_s            xsd:double
  hasIntervalTime_s          xsd:double      S23{2}
  hasTemperature_C           xsd:double
  hasEnduranceCycleCount     xsd:long        renamed from hasCycleCount, which was
                                             also declared on Unit_Process with a
                                             different range and a different meaning
```

```
hasBakeTime_min            xsd:double      S24{3.3}
hasWaveform                { PUND, NDPU, triangle, square, trapezoidal }
hasVthConvention           { linear_extrapolation, constant_current }
                                           S10{II A} vs S11{II}
hasPolarisationStateType   { SS, NSS, OS }  S24{3.3}
fractionMethod             { GIXRD_intensity_ratio, PED_mapping,
                             k_value_inference }        moved here from Phase_State
phasesLumped               { "o+t", "none" }            moved here from Phase_State
```

4.2 Unit_Process

Property	Type	Verified range	Source
hasProcessTemperature_C	xsd:double	150–1050	S1, S9–S11
hasProcessDuration_s	xsd:double	1 to 14400	S10, S24
hasProcessPressure_mbar	xsd:double	7×10^{-4} to 2.7×10^{-1} (PLD); 1.2×10^{-2} to 5×10^{-2} (sputter)	S1 Tables III, IV
hasBasePressure_mbar	xsd:double	see S1 Table III	S1
hasGasFlow_sccm	xsd:double		S1{II B 2}
hasPulseTime_s	xsd:double	0.1, 0.5	S24{2}
hasPurgeTime_s	xsd:double	12	S24{2}
hasOxidantDoseTime_s	xsd:double	optimum 1–5	S1{II A 2}
hasDepositionCycleCount	xsd:int	65 supercycles for ∼10 nm HZO	S24{2}
hasStepIndex	xsd:int	1 to 10 across the eight flows	—
hasGrowthPerCycle_nm	xsd:double	< 0.05 for Cp precursors	S1{II A 1}
hasCoolingRate_Cps	xsd:double	4 and 10 compared	S1{I C}
hasImplantEnergy_keV	xsd:double	40, 50, 60	S10, S11
hasImplantDose_cm2	xsd:double	8×10^{12}, 2×10^{15}, 4×10^{15}	S10, S11
isCrystallising	xsd:boolean	true for E5, an ALD deposition	S24{2}

`hasStepIndex` is a convenience key, not the authoritative ordering. `belongsToFlow` together with `directlyPrecedes` is what the model asserts; the index is a denormalised copy of the same fact, useful for sorting and for spotting a gap in a transcribed flow. It was previously listed among the object properties with no domain and no range.

4.3 Layer_Slot

One slot is one position in one stack. Everything that varies between stacks while the material stays the same lives here.

Property	Type	Verified values	Source
layerIndex	xsd:int	1 at the bottom, ascending upward	—
hasSlotThickness_nm	xsd:double	0.7, 2, 9, 9.5, 10, 20 (ferroelectric); 3–100 (electrodes)	S9–S12a, S23, S24, S27

hasSlotThickness_nm was hasFilmThickness_nm on Material. It moved because the same material individual appears in stacks of different thickness: HZO at 9 nm in Flow B1 S10 and 9.5 nm in Flow G S27. A thickness measured rather than specified — an XRR or TEM figure — is a Measurement on an Artifact, not this property.

4.4 Material

Specified and nominal values only, per §1.3. These describe what the material is or was targeted to be, independent of any one specimen.

Property	Type	Verified values	Source
hasNominalHfZrRatio	xsd:double	1:1 nominal	S9{p.2}
hasDopantConcentration_atPct	xsd:double	La 3.9, 4.5, 7.2; N 0.34 works, 2.44 kills FE	S27{p.1134}, S1 Table III
hasDopantConcentration_molPct	xsd:double	Y 1.9 (sputter), 3.8–5 (PLD), ∼5 (CSD), 5.2 (CSD max Pr)	S1{II B 1, II D}
hasDopantRatio	xsd:string	Al:Hf = 1:30	S9{p.2}
hasBandgap_eV	xsd:double	> 5; 5.6	S1{I}, S23{1}

Moved to Measurement

Everything below was a data property on Material in v2. Each is a characterisation result on a specimen, and each becomes a Measurement with a Measured_Property, a measuredOn artifact and its conditions.

Was	Becomes measuresProperty	Why it cannot stay on Material
hasTrapDensity_cm2	trapped_charge_density	S10 reports 1×10^{14} before cycling and 3.6×10^{14} after, on one specimen S10{III D}. Two values, one material: as a data property the second overwrites the first and the cycling history is lost. The instance document already models these correctly as two Measurement instances; this row is what made the two documents disagree
hasFilmThickness_nm	film_thickness (when measured)	Specified thickness moved to Layer_Slot above; a metrology figure is a measurement
hasCarbonImpurity_atPct	carbon_impurity	TEMA 3.9, TDMA 2.3 S1{II A 1} are values for films grown from those precursors under stated conditions, not properties of the precursor
hasFilmDensity_gcm3	film_density	6.5–9.3 (TEMA), 8.5–12.8 (Cp) S1{II A 1} — ranges across process conditions, not a substance constant
hasGrainLength_nm / hasGrainWidth_nm	grain_size, grain_aspect	PED measurements on S27's specimen: HZO 12–30 × ∼9; TiN bottom 6–15 equiaxed; TiN top 8–15 wide × ∼30 tall S27{p.1134}

Was	Becomes measuresProperty	Why it cannot stay on Material
`hasHfZrRatio` (measured branch)	`composition_ratio`	S24 measured Hf:Zr ∼0.8 on its own film `S24{2}` while calling it HZO. The nominal 1:1 stays on `Material`; the measured 0.8 is a measurement. Keeping both in one property was what made the 1:1 / 0.8 discrepancy look like a contradiction rather than two different claims
`hasCrystallisationOnset_C`	`crystallisation_onset`	HZO from 300, HAO above 650 `S9{p.3}` — read off in-situ from S9's films

Add `composition_ratio` and `crystallisation_onset` to `Structural_Property` under `Measured_Property`.

4.5 Phase_State and phase fraction

Property	Type	Notes
`hasSpaceGroup`	xsd:string	$Pca2_1$, $P4_2/nmc$, $P2_1/c$, Fm-3m, R3m `S1{I A}`
`isFerroelectric`	xsd:boolean	true only for $Pca2_1$ (and ⇒ **UNCERTAIN** contested R3m)
`hasRelativeEnergy_meVperFU`	xsd:double	o is +62 above m in pure HfO_2 `S1{I B}`; Vo lower o by 15 and t by 24 `S1{I B}`
`hasTransformationBarrier_meVperFU`	xsd:double	t→m 223–263; t→o 22–31 `S1{I C}`
`hasPermittivity_k`	xsd:double	⇒ **UNCERTAIN DISPUTED.** S24 (citing Park 2013): m 16–20, o ∼30, t 35–40. S23 (citing Park 2016): o 25–30, t 35–40. Both values are secondary citations from the same research group's work; neither S23 nor S24 measured phase permittivities directly. The t-phase value (35–40) is common to both

`hasPhaseFraction`, `fractionMethod` and `phasesLumped` have been removed from this table. A phase fraction is a measurement: it is produced by a named method on a named specimen, it needs its lumping convention recorded alongside it, and the same film yields different fractions under GIXRD and PED. All three now live on `Measurement` (§4.1), with `phase_fraction` as the `Measured_Property`. §8.1 is unchanged and is the argument for the move.

The four properties that remain are literature values for a phase as such — space group, relative energy, transformation barrier, permittivity — not characterisations of any one film. `hasPermittivity_k` keeps its ⇒ **UNCERTAIN** marking and `uncertaintyRef = U8`.

4.6 Artifact

Property	Type	Verified values	Source
`hasSpecimenLabel`	xsd:string	the source's own name for the sample	—
`hasCyclingHistory_cycles`	xsd:long	0 (pristine) to 10^{10}	S24, S10

An `Artifact` is the physical thing measured, and it is what makes two measurements comparable or not. S12a measures a 1C capacitor and a 1T-1C array cut from the same wafer and reports that they agree `S12a{¶5}` — a claim that can only be stated if the two specimens are distinct individuals. `hasCyclingHistory_cycles` is what separates S10's pristine and cycled trap-density figures without either overwriting the other.

4.7 Memory_Cell_Architecture

Property	Type	Verified values	Source
hasCellArea_um2	xsd:double	0.009 (TSMC OS FeFET) ⇒ **UNCERTAIN** trade press	S6
hasCapacitorDiameter_um	xsd:double	8, 10, 12, 16	S12a{¶4}
hasContactPadArea_cm2	xsd:double	$\approx 2 \times 10^{-5}$	S24{2}
hasGateLength_um / hasGateWidth_um	xsd:double	5/150; 5/150	S10, S11
hasArraySize_bit	xsd:int	16384 (16 kbit)	S12a
hasBitlineLength_bit	xsd:int	32, 64, 128, 256	S12a{¶6}
hasWriteVoltage_V	xsd:double	$+4/-3.5$ (MFIS), $+5.5/-4$ (MIFIS)	S11{III}
hasReferenceVoltage_V	xsd:double	1.8	S12a{¶6}

5 Verified process flows as ordered individuals

Flow	Steps in order	Anneal position	Source
A LabMFM	clean+BE → FE → TE → **anneal 400–1000 °C / 1–60 s** → patterned metal → SC1 etch	before patterning	S1{II, Fig. 3}
B1 GateLastFe-FET	implant → activation 1050 °C/5 s → dHF → IL (O_3, 300 °C) → HZO ALD 300 °C → TiN+W → **anneal 550 °C/60 s** → gate formation → contacts → FGA 450 °C	before patterning	S10{II A}
B2 GateLastFe-FET_MIFIS	B implant → S/D litho + As implant → activation 1050 °C/5 s → dHF → IL (O_3, 300 °C) → HZO (+Al_2O_3) ALD 300 °C → TiN+W → **anneal 400 °C/60 s** → FGA 450 °C	before patterning	S11{II}
C BEOL_FeRAM	FEOL → M1–M6 → TaN BE → ALD HZO 20 nm → TaN TE → **patterning and etch** → **anneal 500 °C** → isolation → V6 via → M7	**after patterning**	S12a{Fig. 1a}
D MFIS_Capacitor	native oxide removal → chemical oxidation → NH_3 RTP + O_2/N_2 → ALD HfO_2 250 °C → TiN PVD → **RTP 800 °C/20 s**	before patterning (none described)	S9{p.2}
E Ful-lALD_AutoAnneal	SiO_2/W → ALD TiN BE 400 °C → ALD HZO 240 °C → [REALD Ru] → **ALD TiN TE 400 °C ≈4 h, crystallises** → litho → plasma etch → Al pads	**no anneal step**	S24{2}
F MFM	pre-clean → ALD TiN BE 400 °C → ALD HZO 270 °C → ALD TiN TE + W → **anneal 500 °C/60 s**	before patterning (none described)	S23{2}
G LaHZO_MIM	Si → ALD TiN 10 nm → ALD La:HZO 9.5 nm → ALD TiN 30 nm → **PMA 673 K, N_2, 1 h**	before patterning (none described)	S27{p.1134}

Derived constraints:

- `CrystallisationTemperature` $\in [400, 800]$ °C across verified flows
- `CrystallisationDuration` $\in [20\text{ s}, 4\text{ h}]$ — three orders of magnitude
- Only Flow C inverts the patterning/anneal order
- Flows D, F, G describe no patterning step at all, which is a lack of evidence

6 Uncertainty register

6.1 Phase fraction is not directly measurable

The orthorhombic and tetragonal peaks overlap or sit very close in GI-XRD and SAED, making phase identification and quantification challenging. `S27{p.1134, citing Chang et al., Acta Mater. 246 (2023)}` The o111/t011 reflection sits near 30.3° 2θ. `S1{I C}`

Consequence: any `hasPhaseFraction` assertion needs `fractionMethod` and `phasesLumped`. The defensible practice is S24's — quantify m only, lump o+t, then infer the split from the C-V permittivity `S24{3.1}`. But the permittivity calibration is itself disputed. S9 does report normalised o(111), o{200} and m(−111) intensities separately `S9{p.3}`, which is the practice S24 warns against.

Also: `hasPhase` on a film is a simplification. Films are polycrystalline mixtures; PED mapping resolves phase grain by grain `S27{Fig. 1a}`. Decide knowingly whether phase attaches to film, grain, or region.

6.2 Contested, conflicting, or implied

#	Item	Status
U1	**Wake-up mechanism**	S24 adopts Mehmood et al.'s two-step framework; its own contribution is showing T_{amb} dependence of $2P_r$ and leakage data that separates V_o redistribution (no leakage rise) from V_o generation (leakage rises at 10^5 cycles). S24's chain: V_o redistribution and generation shift t/o relative stability → t→o transformation → wake-up `S24{3.2}`. S23 argues V_o migration reduces $E_{built\text{-}in}$ directly with no phase change, on the grounds that ε_r stays at ≈ 32.5 `S23{3.2}`. The ε_r datasets are **not directly comparable**: S23's series spans 10^0–10^3 cycles; S24 locates the first wake-up stage between 10 and 10^5–10^6 cycles and measures k only at 10^5 and 10^8. A third mechanism, ferroelastic switching, is untested and cannot be excluded `S24{3.2}`. A fourth, interfacial-layer degradation, is ruled out by S23 and judged unlikely by S24. Stack differences that may explain the divergence: oxidant (H_2O vs plasma O), T_{dep} (240 vs 270 °C), and crystallisation (400 °C/≈4 h vs 500 °C/60 s)
U2	**Fatigue mechanism**	Four candidates. Interfacial screening layer **ruled out**; seed inhibition **unlikely**; domain wall pinning and phase transformation both remain plausible `S24{1, 3.2}`
U3	**Fatigue recovery by heating**	Huang et al. report recovery by mild heating. S24 baked a fatigued Ru device at 105 °C for 1 s / 1 min / 1 h and found **none** — $2P_r$ fell further `S24{3.2}`
U4	**Trapped charge trend during Fe-FET fatigue**	S10 measures it **increasing** 17 → 80 µC/cm^2; Ichihara et al. (Kioxia) measured it **decreasing**. S10 attributes the difference to QSCV overestimating ΔQ_{Si} by 71.1% `S10{I, II C}`
U5	**Curie temperature**	Shiraishi ∼400–500 °C; Hsain's in-situ XRD shows the peak shift only from ∼800 °C. Not reconciled; also thickness- and composition-dependent `S1{I C}`
U6	**Seed layer direction**	Park: a ZrO_2 starting layer degrades performance. Onaya: a bottom ZrO_2 layer gives *higher* P_r than no seed. S1 flags this as an open discrepancy `S1{II A 3}`
U7	**"Antiferroelectric"**	S1 insists on "antiferroelectric-like" because the origin is unproven `S1{I}`; S27 uses the bare term `S27{p.1134}`
U8	**Permittivity calibration for o and t phases**	Both values are secondary citations of Park-group work (S24 cites Park 2013; S23 cites Park 2016). Neither source measured phase permittivities directly. S24 gives o ∼30, t 35–40. S23 gives o 25–30, t 35–40. The t-phase figure (35–40) is common. The difference is a **variation, not a fundamental conflict**
U9	**BEOL thermal ceiling**	S24 asserts 400 °C as the BEOL-compatible figure `S24{1}`; S12a and S23 both anneal at 500 °C `S12a{¶4}`, `S23{2}`; S6 quotes < 350 °C for oxide-semiconductor deposition. **Not a fixed constraint**

#	Item	Status
U10	**Oxygen polarity rule is method-specific**	Lower oxygen favours the ferroelectric phase in ALD and sputter; **PLD is the opposite** — more O_2 pressure gives more o-phase and higher P_r `S1{II C}`
U11	**Phase pathway is method-specific**	ALD/sputter: amorphous → t → o on cooling, irreversible → m on further heating. PLD/epitaxial: **m → t → o on heating** `S1{II C}`
U12	**Thickness ceiling is method-specific**	ALD/sputter tens of nm; PLD to 930 nm; CSD to 1 µm `S1{II C, II D}`
U13	**La doping does not guarantee the o-phase**	At 7.2 at.% La with a 400 °C / 1 h PMA, PED shows the majority phases are **tetragonal and monoclinic** `S27{p.1134}`. The in-situ biasing experiment stated in S27 was "in progress" at time of publication — no results are available
U14	**Electrode substitution is multi-objective**	Ru vs TiN: endurance **improves** ($10^9 \rightarrow\geq 10^{10}$) while fatigue becomes severe ($2P_r \rightarrow 7$ µC/cm^2 at 10^{10}), m-phase rises 22 → 38%, OS retention improves slightly, SS retention worsens. No dominant option `S24{3.2, 3.3, 4}`
U15	**Imprint's effect on P_r is waveform-dependent**	Visible under NDPU, absent under PUND because PUND saturates the polarisation above 3 MV/cm `S23{3.1, 4}`
U16	**V_{th} conventions differ**	Linear extrapolation `S10{II A}` vs constant current `S11{II}`, both on a B1500A. Memory windows are not directly comparable
U17	**Templating by the electrode**	t-phase grains within $< 5°$ of both TiN layers suggest near-epitaxial growth during PMA — S27 says "may be some kind of" `S27{p.1134}`
U18	**Wake-up as a production step**	Verified only as an experimental preparation `S24{3.2}`. v1's claim that it is a manufacturing step rested on an unobtainable source
U19	**FTJ**	Named as an application `S1{I}`, `S27{p.1134}`. No structure or data verified
U20	**S12a's headline device numbers**	30 ns switching, $> 10^4$ s retention, $> 10^{11}$ capacitor cycles appear in the Conclusions with supporting data in Appendices A–D, which are not available. Only the array figure ($> 10^9$ at 3.5 V, 1 µs) has verified conditions
U21	**Internal inconsistencies in verified sources**	S1{II A 2} ozone/water P_r figures and ozone dose figures conflict with S1 Table I; S1 Fig. 3 caption conflicts with the body on step 5; S11 abstract says 10^5 cycles, body says 7×10^4; S23 abstract says 10 s max interval, Fig. 2 caption says 30 s.

6.3 Absences to preserve

Do not let a closed-world reading turn these into negative facts:

- Flows D, F, G describe no patterning step. They were capacitor studies; patterning may simply be undescribed.
- No verified source describes packaging, wafer-level burn-in, or ellipsometry/XRR/AFM/SEM metrology. v1 asserted these; they are removed rather than negated.
- No verified source names a lithography exposure tool.

7 Revision notes for v3

7.1 Internal corrections

These were found by reading the specification against its own instance document. They are errors in v2, not consequences of anything external.

- `hasValue` **renamed to** `hasMeasuredValue`. `owl:hasValue` is a reserved construct.
- `hasCycleCount` **split** into `hasEnduranceCycleCount` (`Measurement`, xsd:long) and `hasDepositionCycleCount` (`Unit_Process`, xsd:int). One name, two ranges and two unrelated meanings.
- **Trap density reconciled.** v2 declared `hasTrapDensity_cm2` on `Material` while the instance document modelled the same S10 figures as two `Measurement` instances. The instance document was right. Resolved by the measured-versus-specified rule in §1.3, which also moved six other properties off `Material`.
- `Artifact` **and** `Technology_Node` **declared.** Both were used as ranges without existing in the class tree.
- `hasTopElectrode`, `hasBottomElectrode`, `layerIndex` **declared.** All three were referenced only inside Notes columns.
- `hasStepIndex` **moved** from object properties, where it had neither domain nor range, to a data property on `Unit_Process`.
- **Ordering facts restated at individual level.** §3.1's bullets named classes; asserting `precedes` between classes would require punning and is incoherent given that Flow C reverses the same two class-level steps.
- `precludes` **added.** `LineOfSight` blocking the 3D capacitor was stated in prose on `enables` and had no property to carry it.

7.2 Changes prompted by the other workstream models

- `Degradation_Phenomenon` **split into** `Film_Level_Degradation` **and** `Array_Level_Degradation`. Prompted by the `Downtime` superclass over `PM` and `Breakdown` in the SMT2020 model, which showed a flat sibling set hiding a real distinction. Here the distinction is what the phenomenon attaches to, and making it explicit exposed that `exhibits` had no domain capable of carrying `Pass_Disturb`. Domain widened to include `Array_Architecture`.
- `Characterisation` **added as a** `Unit_Process` **subclass.** Prompted by the `Quality_Check` node placed inside the process chain in the other FEM model. This specification had `Measurement_Method` and `Measurement` sitting beside the flow rather than in it, which left `Metrology_Tool` unreachable: `usesEquipment` has domain `Unit_Process` and no `Unit_Process` subclass performed measurement. A characterisation step now sits in the flow ordering and produces its measurements through `producesMeasurement`.
- **Scope note on** `Ferroelectric_Material`. The other FEM model covers perovskite and multiferroic ferroelectrics. This one is HfO_2-family by design. The exclusion is now recorded as a `scopeNote` rather than left implicit. The class keeps its general name so the scope can widen later without a rename.
- **Conventions written down (§1).** Naming, annotation properties, disjointness and the open-world reading were all decided but unstated. Both other models state their class definitions in the file itself; this specification kept its reasoning in prose that the OWL file would not carry.

7.3 Found during formalisation

Two conflicts surfaced only when the specification was executed as OWL. Both are recorded in full in `DECISIONS.md` alongside the generator.

- `precedes` **cannot be transitive and asymmetric at once.** The combination is illegal in OWL 2 DL, and a reasoner refuses to load the ontology rather than reporting an inconsistency. Transitivity is kept, since the transitive closure is what derives "the anneal precedes the patterning" from a step chain. Acyclicity is enforced by construction and belongs in SHACL. `directlyPrecedes` is simple and keeps both characteristics.

- **Phases, defects and measurement methods are individuals, not classes.** The class tree in §2 draws them as subclasses; the instance document types them as individuals. The causal properties settle it — `stabilisesPhase` relates `Oxygen_Vacancy` to `Orthorhombic_Pca21`, and object properties relate individuals. The tree notation for these entries should move from `+-` to braces. The subclasses of `Degradation_Phenomenon` are the one group left as classes, to preserve the film-level / array-level split; that inconsistency is recorded rather than resolved.

7.4 Structural change

`Layer_Slot` **reified.** v2 left an open choice between `hasLayerAtPosition` and an RDF list. Closing it in favour of a reified slot resolves three separate problems at once — positional role, one material in two roles across flows, and per-stack thickness — and dissolves the `Si_substrate` dual-typing question flagged at the end of the instance document. Argued in full at the end of §3.3.

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8 Sources

All ten were read in full or near-full. Nothing in this specification derives from a search snippet, an abstract, or a source not listed here.

Tag	Citation	Extent
S1	Hsain, Lee, Materano, Mittmann, Payne, Mikolajick, Schroeder, Parsons, Jones. *Many routes to ferroelectric HfO_2: A review of current deposition methods.* J. Vac. Sci. Technol. A **40**(1), 010803 (2022). doi 10.1116/6.0001317	Full text: I, I A–C, II, II A 1–3, II B 1–3, II C, II D, II E, III; Tables I–V; Figs. 1–21
S5	Duan, Khan, Gong, Narayanan, Ni. *A Full Spectrum of 3D Ferroelectric Memory Architectures Shaped by Polarization Sensing.* arXiv:2504.09713 (2025)	Abstract → "Parallel Stacking 3D Integration"
S6	LaPedus, M. *Next-Gen Ferroelectric Memory: Still A Work In Progress.* Semiecosystem, 8 Feb 2026	Full text. Reports IEDM Dec 2025 papers not read directly
S9	Yang, Lehninger, Reinig, Schöne, Hoffmann, Seidel, Lederer, Gerlach. *Enhanced Performance of FeFET Gate Stack via Heterogeneously co-doped Ferroelectric HfO_2 Films.* arXiv:2508.16768 (2025). Fraunhofer IPMS CNT / TU Dresden	Full text, 6 pp. **MFIS capacitors, not FeFETs**
S10	Zhao, Tian, Xu, Xiang, Li, Chai, Duan, Han, Wang, Wang, Ye. *Experimental Extraction and Simulation of Charge Trapping during Endurance of FeFET with TiN/HfZrO/SiO_2/Si (MFIS) Gate Structure.* arXiv:2106.15939 (2021). IME, Chinese Academy of Sciences	Full text
S11	Hu, Sun, Bai, Jia, Dai, Li, Han, Ding, Fan, Zhao, Chai, Xu, Si, Wang, Wang. *Enlargement of Memory Window of Si Channel FeFET by Inserting Al_2O_3 Interlayer on Ferroelectric $Hf_{0.5}Zr_{0.5}O_2$.* arXiv:2312.16829 (2023). IME CAS / SJTU	Full text, 4 pp.
S12a	Xiao, Peng, Liu, Duan, Bai, Yu, Ren, Yu, Han, Hao. *$Hf_{0.5}Zr_{0.5}O_2$ 1T–1C FeRAM arrays with excellent endurance performance for embedded memory.* Sci. China Inf. Sci. (2022). doi 10.1007/s11432-022-3469-5	Full 2-page Letter. **Appendices A–D unavailable**
S23	Ding, Weng, Lan, Yan, Cai, Qu, Zhao. *Wake-Up and Imprint Effects in Hafnium Oxide-Based Ferroelectric Capacitors during Cycling with Different Interval Times.* Electronics **13**, 1021 (2024). doi 10.3390/electronics13061021	Full text

Tag	Citation	Extent
S24	Khakimov, Chernikova, Koroleva, Markeev. *On the Reliability of HZO-Based Ferroelectric Capacitors: The Cases of Ru and TiN Electrodes.* Nanomaterials **12**, 3059 (2022). doi 10.3390/nano12173059. MIPT	Full text, 16 pp.
S27	Parida, Favia, Richard, De, Popovici, Grieten. *Study of Ferroelectric Phases in $HfZrO_x$ Thin Films with Varying La Content Using Precession Electron Diffraction & In-Situ Heating-Biasing Techniques.* Microsc. Microanal. **31**(7), 1134–1135 (2025). doi 10.1093/mam/ozaf048.588. IMEC	Full 2-page abstract. **Key in-situ biasing experiment stated as "in progress"**